

vARtebrae: Medical Simulation for Spinal Mobilisation Employing Mechanical Metamaterials and Extended Reality

Kiichiro Tatsuzawa*

School of Computing and Information Systems

The University of Melbourne
Melbourne, VIC, Australia

ktatsuzawa@student.unimelb.edu.au

Jiayi Wu*

School of Computing and Information Systems

The University of Melbourne
Melbourne, VIC, Australia

jiayi.wu@unimelb.edu.au

Jo Eann Chong*

School of Computing and Information Systems

The University of Melbourne
Melbourne, VIC, Australia

joeannc@student.unimelb.edu.au

Wenyuan Wu*

School of Computing and Information Systems

The University of Melbourne
Melbourne, VIC, Australia

wenyuanw2@student.unimelb.edu.au

Qiting Ma*

School of Computing and Information Systems

The University of Melbourne
Melbourne, VIC, Australia

qitingm@student.unimelb.edu.au

David Kelly

Department of Physiotherapy
The University of Melbourne

Melbourne, VIC, Australia
d.kelly@unimelb.edu.au

Jessica Stander

Department of Physiotherapy
The University of Melbourne

Melbourne, VIC, Australia

jessica.stander@unimelb.edu.au

Alireza Mohammadi

Department of Mechanical Engineering

The University of Melbourne
Melbourne, VIC, Australia

alireza.mohammadi@unimelb.edu.au

Arzoo Atiq

School of Computing and Information Systems

The University of Melbourne
Melbourne, VIC, Australia

arzoo.atiq@unimelb.edu.au

D. Antony Chacon[†]

Faculty of Engineering and IT
The University of Melbourne

Melbourne, VIC, Australia

antony.chacon@unimelb.edu.au

Abstract

Spinal mobilisation education traditionally relies on simulated patient-based learning approaches, which face challenges such as the absence of objective feedback mechanisms, the risk of harm to volunteers, limited scalability, and lengthy learning processes. To overcome these limitations, we propose *vARtebrae*—a tangible user interface (TUI) that incorporates 3D-printed metamaterials and a silicone layer to replicate the physical fidelity of the human lower back anatomy. *vARtebrae* features a magnetometer-based sensor system capable of accurately measuring deformation and providing objective visual feedback via an extended reality (XR) app. The app offers real-time visualisation of spinal movements, colour-coded force cues, and a force-time graph to support the

development of manual motor techniques. Our design process followed a user-centred approach involving physiotherapy experts. Early feedback suggests that *vARtebrae* can improve training by addressing limitations in traditional methods. With its scalable, cost-effective, and student-centred design, *vARtebrae* represents a significant advancement in spinal mobilisation training.

CCS Concepts

- **Human-centered computing** → **Haptic devices; Mixed / augmented reality; Information visualization; User centered design;**
- **Hardware** → *Simulation and emulation.*

Keywords

Mechanical Metamaterials, Extended Reality, Medical Simulator, Visuo-Haptic Feedback, Physiotherapy Education

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*These authors contributed equally to this work.

[†]Corresponding author

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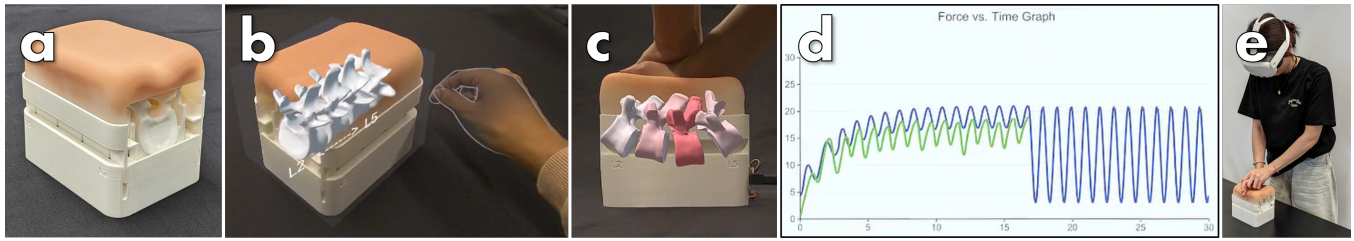


Figure 1: vARtebrae system overview: a) physical device to provide haptic feedback, b) calibration for virtual-physical vertebrae alignment in XR app, c) visualisation of bones movement and colour tones based on force exerted, d) force-time graph showing expert benchmark (blue) and user performance (green), and e) a user in a Meta Quest 3 headset interacting with the system.

1 Introduction

The spine is a vital chain that orchestrates our every movement; however, like any chain, it is only as strong as its weakest link. For many, this weak link manifests as back pain—a common musculoskeletal condition that can severely impact daily life. Clinical approaches such as spinal mobilisation aim to relieve this condition, but they require precise, repetitive, and well-practised motor skills to ensure safe and effective treatment [29]. Traditionally, learning these techniques rely on volunteers, presenting logistical challenges and risks of injury due to inexperience. This limits students' opportunities for extensive, high-quality practice. When practice sessions do occur, students often lack effective feedback mechanisms [7]. Instructors face the difficult task of translating students' momentary physical inputs into clear, actionable verbal guidance—feedback that is not only challenging to deliver but also difficult for students to interpret and apply effectively. Although some tools aim to address these issues, current technologies remain limited.

There is a clear demand for an integrated solution that is haptically authentic, cost-effective, scalable, and capable of enhancing both the quality and duration of practice. Addressing these challenges, this paper introduces *vARtebrae*, an educational tangible user interface (TUI) with two key contributions: (1) the design and integration of mechanical metamaterials to simulate soft tissues in medical simulators, and (2) an extended reality (XR) app providing real-time 3D visual feedback for manual motor techniques.

2 Related Work

2.1 Approach and Technologies in Spinal Manual Therapy Education

Spinal manual therapy, commonly used to address mechanical spinal issues, involves precise mobilisation/manipulation of spinal joints to alleviate pain through physiological, nervous, and muscular changes [10, 20]. Posterior-anterior lumbar mobilisation, for instance, applies controlled force to the vertebrae to relieve lower back pain. Traditionally, this therapy is taught through a clinical model where students practice hands-on techniques with peers or real patients, relying on in-class demonstrations and verbal instructor feedback [23]. However, this approach has challenges: it lacks objective feedback for evaluating performance, poses safety risks for peers and volunteers, and is resource-intensive, limiting scalability and consistency [29].

Technologies in spinal mobilisation education address specific needs but have limitations. Manikins and skeletal models offer a platform for practising techniques but can't replicate real spinal conditions, such as stiffness or injury. They also lack real-time feedback on the force exerted, hindering skill refinement [9]. Force-sensing technologies, such as force-monitoring tables, provide insights into pressure applied during spinal mobilisation but fail to simulate the haptic response of human tissue, limiting their realism [8, 31]. Treatment tables with embedded sensors allow practice with manikins or human volunteers, but manikins lack haptic fidelity, and human volunteers pose risks of harm and scalability issues. Similarly, skin-mounted sensors on manikins or participants are more affordable but face the same limitations [9]. These technologies, while useful for foundational training, struggle to replicate real-world scenarios and offer limited unsupervised practice. These challenges highlight the need for a scalable, student-centred, and cost-effective solution that integrates theoretical learning with realistic hands-on experience, providing objective feedback, improved safety, and streamlined training to enhance spinal manual therapy education.

2.2 Tangible User Interfaces for Spinal Mobilisation Training

Tangible user interfaces (TUIs) allow users to interact with digital information through the physical manipulation of objects. In this context, medical simulators serve as a form of educational TUIs, helping to develop manual and procedural skills. In spinal mobilisation training, such simulators aim to replicate the mechanical properties and haptic feedback of a human spine, ensuring physical fidelity to real-world conditions for improved skill transfer. One example is *SpinalLog* [6], which combines a physical lumbar spine model with sensors to measure vertebral displacements and applied forces, alongside a graphical user interface (GUI) that provides real-time visual feedback.

While *SpinalLog* eliminates the need for human volunteers, offers realistic deformation, and provides visual feedback on exerted forces, it also has notable limitations. First, its sensor system performs poorly. *SpinalLog* relies on conductive foam, which degrades under prolonged and repetitive deformation, reducing reliability and requiring frequent recalibration. This affects its accuracy and usability over time. Second, it lacks back soft tissue emulation. Without soft tissue above the vertebrae, the simulator lacks the fidelity needed to closely replicate the feel of a real human back. These

shortcomings hinder the effectiveness of *SpinalLog* in delivering a fully immersive and realistic training experience.

Recent advances in material science and digital fabrication techniques offer cost-effective solutions to address these shortcomings [28]. In particular, mechanical metamaterials [17] and cast silicone can be combined to produce suitable mediums that mimic human soft tissue, such as muscles and skin. Moreover, mechanical metamaterials can replicate multiple levels of haptic feedback [24], similar to the mechanical behaviour of the foam-based sensor units in *SpinalLog*, but with enhanced durability and reliability.

2.3 Extended Reality in Medical Simulators

Extended reality (XR) technologies, encompassing virtual (VR), augmented (AR), and mixed reality (MR), have gained increasing adoption in medical training [2, 11]. These tools enhance understanding of complex three-dimensional structures and facilitate engagement in medical procedures. MR and AR, in particular, blend digital elements with real physical environments, offering significant potential to improve learning experiences for medical professionals [1, 33]. Research indicates that XR with headsets provides several advantages over traditional learning methods [26]. For example, in gross anatomy education, XR apps are reusable, cost-effective, and eliminate the need for expensive and maintenance-intensive cadaver-based learning [14]. XR is also widely applied in surgical simulations, where, in combination with motion tracking and haptic feedback technologies, create realistic environments for skill practice. Furthermore, XR headsets enable the display of critical information more effectively, reducing distractions compared to traditional monitor-based systems [3]. Lastly, XR provides a safe environment for trainees to practice, minimising risks to patients and allowing learners to make mistakes without fear [14, 19].

Despite these advancements, there is no existing work that focuses on the visualisation of spinal mobilisation for training physiotherapists. The closest area of XR application is in cardiopulmonary resuscitation (CPR) training, which offers a compelling example of XR's potential for physiotherapy education. CPR training integrates manikins equipped with sensors and virtual reality applications to immerse users in realistic scenarios, enhancing the authenticity and effectiveness of the training process [32, 33]. These systems collect real-time data from the manikin's force sensors, capturing the depth and rate of chest compressions, which are then processed and displayed in XR headsets. This setup provides dynamic feedback on users' techniques and helps refine their skills through visual instructions, including 3D demonstrations of proper technique [12]. Such applications demonstrate XR's ability to improve precision and effectiveness in medical training and highlight an opportunity to explore its benefits in spinal mobilisation education, potentially improving the way physiotherapists acquire critical motor skills.

3 vARtebrae

3.1 Design Process

For a spinal manual therapy simulator, two critical factors are the physical fidelity of the spine and the palpability of its motion [6]. To address these needs, we adopted a user-centred design approach [25] to develop a device that accurately replicates spinal movements and tactile sensations. Our design follows the base

structure of the *SpinalLog* [6], which used sensors placed under L-shaped supports to hold each vertebra. Focusing on the lumbar spine, we covered vertebrae L2 to L5, which were identified as essential for practising spinal mobilisation techniques in this region. To provide users with visual feedback, we implemented an XR app that displays the magnitude and location of the forces exerted during these techniques. Our methodology followed four stages: (1) defining system requirements, (2) researching and designing prototypes, (3) iterative optimisation and refinement, and (4) conducting evaluations to enhance usability. Our team includes two physiotherapy experts (36 and 20 years of practice experience, 31 and 18 years in education) who provided continuous feedback on our system's visual and haptic aspects.

3.2 Back Soft Tissues Emulation

Our initial attempt to incorporating soft tissues in the device involved using a top layer made entirely of cast silicone. However, this design severely limited relative movement between the vertebrae. While softer-grade silicone improved inter-vertebral flexibility, the overall haptic feedback was too soft to accurately simulate a human back. To overcome this, we developed a two-layer design mimicking the lumbar spine anatomy, as shown in Figure 2a. Inspired by Ion et al.'s work [17], the design features a 3D-printed metamaterial structure combined with a silicone layer. The metamaterials mimic the muscle layers between vertebrae and the back skin, providing flexibility for intervertebral movement, while the outermost silicone layer replicates human skin. This approach significantly enhanced the simulator's shape and overall haptic fidelity.

The metamaterial layer was designed with a honeycomb pattern oriented perpendicular to the sagittal plane, as depicted in Figure 2c. The hexagons have a vertex-to-vertex diameter of 10 mm and a wall thickness of 1 mm, 3D printed with TPU 95A. The outer silicone layer was cast using a 3D-printed PLA mould, as shown in Figure 3. We used Platsil Gel-0030 prosthetic grade silicone to produce a 5 mm-thick layer replicating human skin. To ensure a smooth surface, the bottom part of the mould was printed laying on its longest side to minimise texture from layer lines, and its interior surface was sanded down to refine the silicone's outer finish.

3.3 Metamaterial-based Sensors

Building on the metamaterials developed by Mohammadi et al. [24], we employed 3D-printed mechanical metamaterials embedded with magnetometers and magnets to precisely measure linear deformation and deliver consistent haptic feedback (Figure 2b). These materials provide greater durability and reliability compared to the conductive foam used in *SpinalLog*, significantly reducing the need for frequent recalibration.

Our design features a spring-shaped, two-level structure fabricated with TPU 95A. This geometry enables dual-level compression, as shown in Figure 2d, designed to replicate the mechanical behaviour of the human back during spinal mobilisation. Each vertebra lies on top of a pair of these structures printed as a single unit with a width of 30 mm. In their resting state, the structures have a height of 25 mm, with a maximum designed compression limit of 13 mm. Through iterative testing with physiotherapy experts, we determined layer thicknesses of 0.8 mm for the upper part and

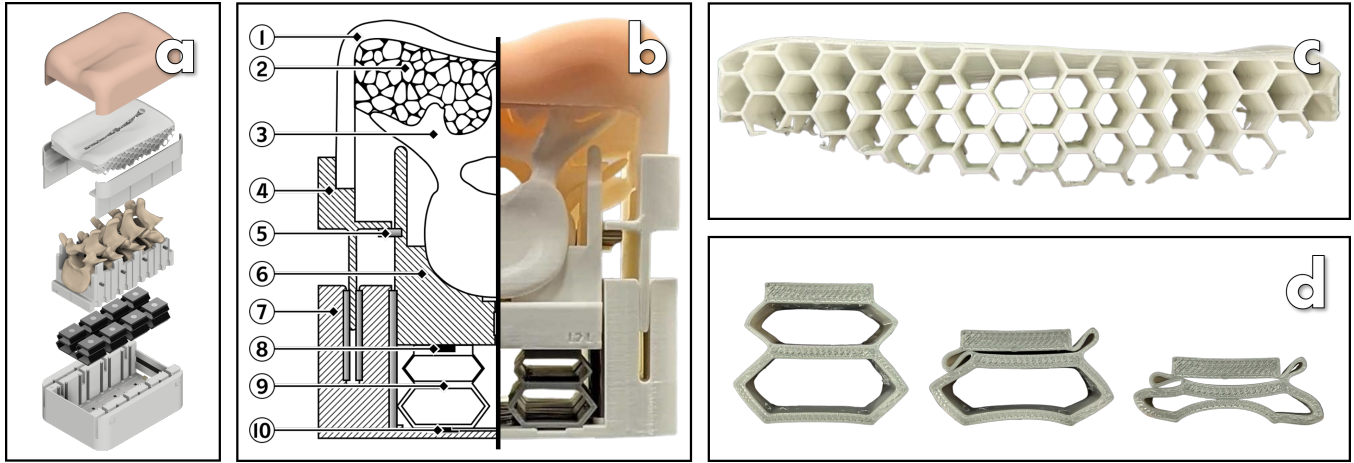


Figure 2: vARtebrae mechanical design. a) Exploded view showing layered design. b) Sectional view of the front, showcasing: (1) cast silicone to emulate skin, (2) metamaterials to emulate muscle, (3) 3D-printed bone, (4) sliding support for the silicone, (5) stainless rods to facilitate parts movement, (6) L-shaped support, (7) base, (8) magnet on top of metamaterial-based sensor, (9) metamaterial-based sensor unit, and (10) magnetometer. c) Sagittal view of honeycomb-shaped metamaterial that emulates muscles on top of the vertebrae. d) Metamaterial-based sensor unit and its two-level deformation.

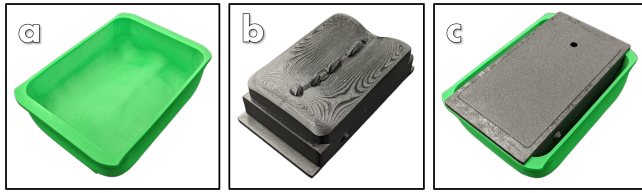


Figure 3: Cast mould used to create the silicone skin: a) bottom part, b) top part, and c) assembled mould.

1.8 mm for the lower part, achieving an overall stiffness in the system slightly greater than that of a healthy back.

Each spring-shaped unit integrates a magnet and a magnetometer to measure vertical linear displacement, resulting in a total of eight sets (two per vertebra). A cylindrical neodymium magnet, with a diameter of 8 mm, height of 3 mm, and nominal pull force of 74 N, is positioned at the top of each unit. An MLX90393 magnetometer is located at the base, as illustrated in Figures 2b and 4b. We gathered the magnetometers readings with an ESP32 Feather V2 board via the I²C protocol. To overcome the MLX90393's limitation of supporting only four unique I²C addresses, we employed a PCA9546A I²C multiplexer, as shown in Figure 4a.

To convert magnetometer readings (magnetic field strength) into linear displacement measurements in millimetres, we previously collected data using 3D-printed spacers with known distances (Figure 4d). Since the magnetic field strength decreases with distance at a cubic rate, we performed a regression analysis to fit a cubic root function that models the relationship between magnetic field strength (independent variable) and distance (dependent variable) (Figure 4c). This equation was then applied to obtain displacement from each sensor unit during our system operation:

$$y = -20.032 \cdot x^{1/3} + 3.995 \cdot x^{1/2} - 0.006 \cdot x + 107.15 \quad (1)$$

3.4 XR App

We utilised the ESP32's Bluetooth capability to wirelessly send sensor data to external devices for visual feedback to the user, such as a mobile device or XR headset. In this paper, we showcase the XR app developed for the Meta Quest 3 in Unity employing Meta XR packages. We chose the Meta Quest 3 for its wide commercial availability and support, and to leverage its passthrough capabilities.

3.4.1 Calibration. In the XR app, users must first generate a virtual twin of the bones corresponding to the physical vARtebrae device by performing a pinch gesture with their right thumb and index finger. This gesture creates a transparent virtual cube at the pinch location, serving as the digital counterpart of the physical device. Users then manually adjust the virtual cube's position to ensure alignment with the physical device, as illustrated in Figure 1b. Since vARtebrae is designed to remain stationary, this manual calibration method does not significantly inconvenience users. The decision to use manual alignment stems from limitations in accessing the Meta Quest 3's front cameras during development due to privacy constraints. These restrictions, at the time of our system's creation, prevented the implementation of automated computer vision-based calibration techniques without incorporating additional components (i.e., external cameras or trackers).

3.4.2 Passthrough bones visualisation. After calibration, users can operate our system by applying force with specific hand and arm postures for spinal mobilisation techniques as seen in Figure 5. As force is applied, the device provides a real-time 3D visualisation of the virtual spine's movements, including downward vertical displacement and rotation in the transverse and sagittal planes, as shown in Figures 1c and 5c. Additionally, we have incorporated colour-coded cues into the visualisation. When users apply excessive force, the bones are highlighted in red tones (Figure 1c). This

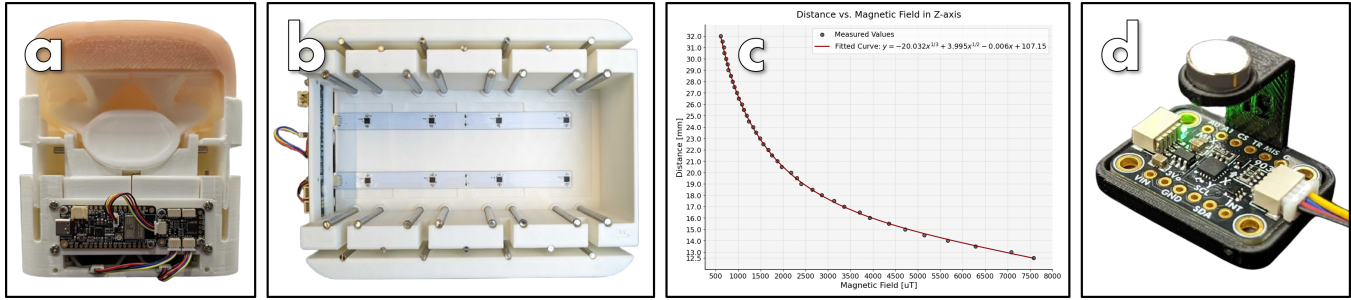


Figure 4: *vARtebrae* electronics overview: a) ESP32 Feather V2 board and PCA9546A I²C multiplexer at the back of the device, b) MLX90393 magnetometers arrangement at the base of the device, c) relationship between magnetometer readings and distance from magnet in Z-axis, and d) sample 3D-printed measuring tool to obtain magnetometer readings at set heights.

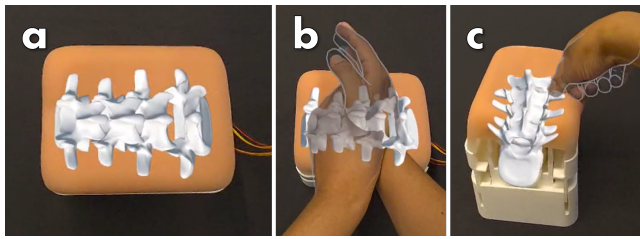


Figure 5: Passthrough virtual bones visualisation from the user's point of view: a) hands-free top view of the system, b) top view when placing hands ready for force exertion, and c) spine rotation when applying some force to a side.

colour-change feature aims to help users understand the appropriate amount of force to apply, enabling them to adjust their technique in real time.

3.4.3 Force-time graph. We adapted the original force-time graph from the *SpinalLog* [6] for use in our XR app as it is deemed critical for learning spinal mobilisation [27]. By aggregating the vertical displacements from all eight sensors, we render the total “force” in real time, which is proportional to the sum of all displacements, as illustrated in Figure 1d. This graph feature allows users to load various force patterns or techniques, helping them refine and practice their skills by comparing their applied force against expert-generated benchmarks.

4 Preliminary Evaluation of XR App

We conducted a user evaluation comprising usability testing and post-task interviews to gather qualitative data on participants' experiences with the *vARtebrae*'s XR app. This study received approval from the Office of Research Ethics and Integrity of the University of Melbourne (Ethics ID 20833).

We recruited 8 participants (3F/5M), aged between 21 and 25 years (mean = 23.75, SD = 1.39), who performed tasks designed to evaluate all major features of the system. Participants first performed manual calibration in the XR app (Figure 1b), then exerted force on the L3 vertebra with their hands as shown in Figures 5b and 5c, while visualising the virtual bones and the force-time

graph. After the tasks, we interviewed participants, assessing usability, user experience, and areas for improvement. Each session lasted approximately 30 minutes.

The qualitative data collected was analysed using affinity diagramming, which revealed the following key themes:

- **User experience and design:** While the XR app was generally well-received, four participants found certain interactions, such as the manual calibration process, challenging. They described aligning the virtual cube with the physical device as unintuitive, leading to suggestions for integrating visual guides or step-by-step instructions. One participant recommended using QR-based automated tracking to simplify the process as seen on other XR systems. Once calibrated, however, most participants found the system straightforward and intuitive to use. Enhanced instructions were recommended to reduce the learning curve for first-time users of XR headsets with haptic devices.
- **Anatomy visualisation:** The visualisation of the virtual vertebrae was highly praised by all participants, who highlighted its value in demonstrating the dynamic movement of spinal structures during force exertion. However, two users suggested fine-tuning the sensitivity of virtual bone rotations to better capture subtle motions, which are essential for teaching precision. By combining tactile and visual feedback, *vARtebrae* was deemed especially useful for novice learners who struggle to understand the vertebral anatomy.
- **Educational tools:** Two participants noted that the force-time graph significantly enhanced the educational experience by reinforcing concepts related to pressure application over time. One participant suggested that it would be helpful to have the ability to reposition the graph in space, as it currently appears directly in front of the user's view. Additionally, two participants recommended adding automated performance analysis and benchmarking tools to support self-directed learning.

5 Discussion and Future Work

Our work highlights the potential of *vARtebrae* as an advanced visuo-haptic training tool for spinal mobilisation, offering significant improvements over previous systems while identifying areas for further enhancement. The iterative process with physiotherapy

experts during the development of the haptic device resulted in a more realistic tactile feedback. This response aligns with earlier works such as *SpinalLog* [6], which emphasised the importance of physical fidelity in spinal models. However, *SpinalLog* faced limitations due to sensor degradation and lack of soft tissue simulation. By integrating mechanical metamaterials and cast silicone, *vARtebrae* addresses these issues, offering a more reliable and anatomically accurate solution. Despite these improvements, achieving perfect haptic replication remains an ongoing endeavour. Future developments could explore different geometries and arrangements of the metamaterial-silicone combination for higher haptic fidelity.

The user experience and design of the XR app received positive feedback in our study. However, the manual calibration process was found to be unintuitive by some participants, a challenge that is consistent with other XR-based systems requiring precise alignment of virtual and physical interfaces [13]. Integrating automated tracking solutions into the physical device, such as the VIVE Ultimate Tracker [16], could streamline the setup process and improve consistency.

Dynamic visualisations, including colour-coded force cues and passthrough bones, were identified as strengths of the system. Similar feedback mechanisms in CPR simulations have demonstrated their effectiveness in improving procedural precision [4]. Nevertheless, some participants recommended fine-tuning the sensitivity of bone rotations to better capture subtle motions, an observation supported by Sheaves et al. [29], who emphasised the importance of precision in teaching manual therapy techniques. Addressing this feedback could enhance the educational value of the system for advanced learners.

The combination of tactile and visual feedback in *vARtebrae* is particularly beneficial for novice learners, emphasising the importance of multimodal feedback in medical training, as highlighted by prior studies [6, 7, 31]. However, unlike XR-based CPR training systems that incorporate automated performance analysis, *vARtebrae* currently relies on subjective evaluations by instructors. Incorporating AI-driven feedback, as suggested by Winkler-Schwartz et al. [34], could allow students to engage in self-directed learning and benchmarking. Moreover, the absence of immersive training menus and audio cues limits its instructional potential. Enhancing these features could help align the system more closely with modern pedagogical approaches in medical education.

Our system currently supports the visualisation of forces exerted, either through colour-changing bones or a force-time graph. However, it does not provide guidance on proper body, arm, or hand postures for applying these forces effectively. Maintaining correct postures is essential for delivering the right forces on the spine and ensuring the practitioner's wellbeing [18]. Future research will focus on integrating arm and hand tracking, either through the XR headset itself [15] or with wearable sensors [5, 22], to assist students in performing physiotherapy techniques with proper postures.

Regarding the technical aspect, the force-displacement data used was largely simplified, relying on a proportional factor to estimate force magnitudes similar to the *SpinalLog* system [6]. We plan to conduct compression testing with our metamaterial-based sensor units to establish a more precise mapping between load and displacement. This will allow us to calibrate the system with real-world data, providing more reliable feedback on force exerted in

the XR app. Another challenge is to simulate the variability in musculoskeletal stiffness. Currently, this is addressed by manually swapping the metamaterial-based sensor units after partially disassembling the device. To enhance this, we intend to incorporate actuation methods, such as pneumatics or cable-driven systems from soft robotics [21, 30, 35, 36], into the sensor units. This would allow real-time adjustment of stiffness, creating a more adaptive and realistic training experience.

Finally, while our preliminary qualitative study on the XR app provides valuable insights, further research is essential to validate the whole system with physiotherapy experts, such as clinicians and educators, and students. Additionally, studying the long-term effect of visual and haptic feedback on skill acquisition and retention is crucial. Monitoring performance over time and conducting in-depth user interviews will offer essential data to evaluate the system's overall effectiveness.

6 Conclusion

We present *vARtebrae*, a novel medical simulator designed for spinal mobilisation education. By integrating mechanical metamaterial and silicone-based soft tissue emulation, magnetometer-based displacement sensors, and extended reality technologies, *vARtebrae* addresses longstanding challenges in physiotherapy training. The system replicates the physical fidelity of human lumbar anatomy while delivering real-time, interactive visual feedback, enabling students to develop manual motor skills more effectively and safely. Early feedback highlights *vARtebrae*'s transformative potential, combining haptic fidelity with XR visualisation to enhance medical training and showcase the versatility of mechanical metamaterials in replicating organic tissue properties. Future work will focus on refining calibration processes, incorporating variable stiffness mechanisms for enhanced adaptability, validating the system with physiotherapy experts and students, and conducting longitudinal studies to evaluate its impact on learning outcomes.

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